

DEVELOPMENT AGREEMENT
MOUNTAIN CEMENT COMPANY, LLC TRANSLOADING FACILITY

THIS AGREEMENT is made this 18th day of August, 2026 (the “Effective Date”), by and between Mountain Cement Company LLC, organized under the laws of the State of Delaware, whose address is 5 Sand Creek Rd, Laramie, Wyoming 82070, hereinafter referred to as “the Applicant,” authorized to do business in the State of Colorado, and the City of Evans, a Colorado home rule municipality whose address is 1100 37th Street, Evans, Colorado 80620, hereinafter referred to as “City.” The Applicant and the City are collectively referred to as the “Parties.”

RECITALS

WHEREAS the Applicant owns the real property located in the City of Evans in Weld County, Colorado identified by Assessor Parcel Number 105703400002 (“the Property”), described in Exhibit A, attached hereto and incorporated by reference; and

WHEREAS, the Applicant has recorded a Site Plan, described in Exhibit B (“Site Plan”), and an application for a transloading terminal (the “Project”) on the Property; and

WHEREAS the Parties agree that the Project will generate an increase in traffic in the City of Evans, as demonstrated in the application materials; and

WHEREAS the Parties agree that City rights-of-way, along the haul/travel routes, described in Exhibit C (“Haul Route”), specifically to 47th Avenue, also known as Weld County Road 33, attached hereto and incorporated by reference, which serve the Property, will require increased maintenance and increase the need for permanent improvements due to the increase in traffic caused by the Project, and

WHEREAS, the Parties desire to enter into this Development Agreement (“Agreement”) for the use of the Haul Route, permanent improvements for the Haul Route, and maintenance of the Haul Route (as hereinafter defined); and

WHEREAS the Project shall not be allowed to commence operation nor receive a Certificate of Occupancy until this Agreement is approved by Evans City Council.

**NOW, THEREFORE, IN CONSIDERATION OF THE FOREGOING AND OF THE PROMISES
AND COVENANTS CONTAINED HEREIN, THE PARTIES HERETO PROMISE, COVENANT,
AND AGREE AS FOLLOWS:**

PART I: RECITALS INCORPORATED, SITE-SPECIFIC PROVISIONS

- A. **Recitals Incorporated.** The Recitals above are expressly incorporated into this Agreement.
- B. **Haul/Travel Routes.** The Applicant shall be financially responsible for its Proportional Share (as hereinafter defined) of the City’s reasonable and documented out-of-pocket costs associated with maintaining and improving the Haul Route, in each case, in accordance with the terms of this Agreement.
- C. **Approved Access.** Haul trucks and traffic shall enter and exit the Property only at the approved access on Weld County Road 33, now known as 47th Avenue and shall remain on determined roadways for further dispersal. Any City roads used by traffic associated with the Project may

become part of the established haul/travel routes as determined by the City Engineer, in their sole reasonable discretion, so long as otherwise in compliance with this Agreement.

- D. **Weight Limits.** Travel vehicles must not exceed CDOT required specifications for pounds per axle and axle configurations unless overweight permits have been applied for and granted in writing.
- E. **Temporary Deviations.** If maintenance of the Haul Route, the Permanent Improvements, or other City of Evans or Weld County projects mandate deviation from the above-mentioned haul/travel route, the City Engineer shall authorize, in writing (e-mail acceptable), the deviation and shall provide alternative access to the Property. In that circumstance, haul/travel vehicles will utilize paved city roads whenever possible. A deviation from the established haul/travel route lasting more than six (6) months or requests for one (1) or more deviations within a twelve (12) month period shall authorize the City Engineer to (i) change the haul/travel routes designated herein, in a commercially reasonable manner, (ii) amend Exhibit C accordingly, and (iii) provide notice to the Applicant, without the need for City Council approval.
- F. **Haul Route Signage.** Applicant shall install travel route signs as per Manual of Uniform Traffic Control Devices standards, at all exit points of the Property which can be clearly seen by drivers leaving the Property and which clearly depict City approved travel routes. Signage shall be installed to prohibit all traffic from accessing the Property from the north.
- G. **No Deviation from Permitted Haul Routes.** Except as provided in Section 1.E or as authorized by the Evans City Council pursuant to this Agreement, no use of any City roads other than those specifically authorized for use in this Agreement to access the Property is permitted without the prior written amendment of this Agreement by City Council.
- H. **Interim Period.** The time period beginning on the Effective Date and ending on the earlier of: (1) the date the Permanent Improvements are completed and the City provides a certificate of completion to Applicant with respect thereto, or (2) the sixty (60) month anniversary of the Effective Date, is considered the "Interim Period," which shall be subject to extension by the Tail Period, as hereinafter defined. Notwithstanding the foregoing, if (i) the Permanent Improvements are not fully funded (i.e., the City has not received full funding therefor, including all necessary funds budgeted and appropriated by the Evans City Council) during the first four (4) years of the Interim Period (i.e., prior to the forty-eight (48) month anniversary of the Effective Date), and (ii) the City receives full funding and has appropriated and budgeted those funds for the Permanent Improvements at any time during the last twelve (12) months of the five-year Interim Period, then the Interim Period shall automatically be extended for an additional six (6) months following the sixty (60) month anniversary of the Effective Date (such additional period, the "Tail Period," and the date on which the Tail Period expires, the "Tail Period Expiration Date"), solely to afford the City additional time to complete construction of the Permanent Improvements. If the Tail Period applies pursuant to the foregoing sentence, all references in this Agreement to the "Interim Period" shall thereafter mean the Interim Period, as extended through the Tail Period Expiration Date. For the avoidance of doubt, the City shall have no right to, and the Tail Period shall not apply, extend, or otherwise take effect, if the City receives full funding for the Permanent Improvements at any time prior to the forty-eight (48) month anniversary of the Effective Date, but fails to complete construction of the Permanent Improvements prior to the expiration of the Interim Period. In such circumstance, this Agreement shall terminate in accordance with Part III Section Q.3 below, and the City shall have no claim or entitlement to the Tail Period or any other extension of the Interim Period.

Part II: Improvement and Maintenance Requirements:

- I. **Immediate Site Access Improvements.** The Applicant shall be responsible for the construction of certain safety improvements, consisting of one streetlight at the Property entrance, construction of adequate turning radii at the Property entrance which extend into City Rights-of-Way ("ROW") and signage as provided in Section 1.F. Any approved access location improvements will include standard City approved tracking control and may include a double cattle guard set (one right after the other), placed back-to-back across the entire width of the access, to ensure a complete revolution of the truck tires in order to minimize the tracking of mud and debris onto the adjacent road. These improvements shall be completed prior to operation of the Project.
- J. **Interim Off-Site Maintenance.** Maintenance of the Haul Route will be undertaken by the City and, during the Interim Period only, Applicant will be responsible for Forty Percent (40%) of the reasonable and documented out-of-pocket maintenance costs incurred by the City in order to maintain the Haul Route in its current and existing condition as a rural local gravel roadway during the Interim Period. The current and existing condition of the Haul Route is evidenced by the descriptions and depictions attached hereto as Exhibit D (the "Current Condition"). Applicant will pay to the City Applicant's proportional share (40%) (such percentage, "Proportional Share") of such maintenance costs within forty-five (45) days of its receipt of the City's invoice therefor, subject to good faith dispute.
- K. **Permanent Improvements.** The City shall develop a design for permanent improvements to the Haul Route, which shall be limited to asphalt paving (after the Haul Route road base is sufficiently built up) and according to the City's published Engineering Standards and Specifications for rural paved roads, which are attached hereto as **Exhibit E** (the "Permanent Improvements"). The City will share its design with the Applicant for review before proceeding with construction of the Permanent Improvements.

As of the Effective Date, the City's good faith estimate of the total cost of the Permanent Improvements, for the purposes of this Agreement, is ONE MILLION FIVE HUNDRED THOUSAND DOLLARS (\$1,500,000.00) (the "Cost Estimate"). Notwithstanding anything contained herein to the contrary, in no event shall the City seek contribution from the Applicant with respect to the Permanent Improvements for more than a dollar amount equal to the product of (x) Applicant's Proportional share, multiplied by (y) the Cost Estimate (such resulting amount, "Applicant's Permanent Improvements Cap"). Subject to the City's receipt of sufficient funding for the Permanent Improvements during the Interim Period, including receipt of available funds budgeted and appropriated by the Evans City Council, the City will lead, coordinate and complete the construction of the Permanent Improvements during the Interim Period as a capital infrastructure project.

Within forty-five (45) days after the Effective Date, Applicant shall contribute its Proportional Share for the Permanent Improvements by electing one of the following methods: (i) paying the full amount of Applicant's Proportional Share of the Cost Estimate in readily-available funds into an escrow account held by a mutually-agreed third party financial institution until substantial completion of the Permanent Improvements occurs; or (ii) providing a payment bond or other surety instrument, at Applicant's election and as reasonably acceptable to the City. In the event the Applicant elects to deliver to the City a payment bond or other surety instrument, such instrument shall be issued by a surety company authorized to do business in the State of Colorado and reasonably acceptable to the City, in an amount equal to One Hundred Fifteen Percent (115%) of the Applicant's Proportional Share of the Cost Estimate for the

Permanent Improvements. The City shall have the right to draw upon the funds in the escrow account, the bond or the surety instrument, as applicable, to fund completion of the Permanent Improvements in an amount equal to (x) Applicant's Proportional Share, multiplied by (y) the actual cost of the Permanent Improvements (which resulting amount shall in no event exceed Applicant's Permanent Improvements Cap); provided, however, if Applicant provides a bond or surety instrument in accordance with this paragraph, the Parties agree that (i) Applicant will pay to the City Applicant's Proportional Share of the reasonable and documented out-of-pocket costs incurred by the City with respect to the Permanent Improvements within forty-five (45) days of its receipt of the City's invoice therefor, subject to good faith dispute, and (ii) the City shall not seek payment from any surety under such bond or instrument unless Applicant wrongfully fails to pay such amounts with respect to the Permanent Improvements under this paragraph after written notice from the City and fails to cure such failure as required by the terms of this Agreement. Any payment bond or surety instrument shall remain in effect until the expiration of the Interim Period or substantial completion of the Permanent Improvements. Following substantial completion of the Permanent Improvements, any remaining escrow funds shall be automatically released by the escrow agent to the Applicant and the bond or surety instrument provided by the Applicant shall be released, in each case, as applicable.

L. Other Maintenance. Off-site recurring maintenance is required during the Interim Period as follows:

1. Need for Immediate Repairs. If damage to a designated haul/travel route caused exclusively by Project traffic causes an immediate threat to public health and safety or renders the road impassable ("Significant Damage"), the City Engineer or their designee, shall, after inspection, notify Applicant in writing (e-mail acceptable) of such Significant Damage. Applicant shall identify the repair required and shall consult with the City Engineer or their designee on the extent, type, timing, materials and quality of repair (i.e., temporary versus permanent) within five (5) business days after receipt of such notice and shall commence such repair within twenty (20) business days after receipt of such notice. If Applicant identifies Significant Damage prior to receiving notice thereof from the City Engineer or their designee, Applicant shall notify the City Engineer in writing (e-mail acceptable) of such Significant Damage. The City Engineer shall consult with Applicant or its designee on the extent, type, timing, materials and quality of repair (i.e., temporary versus permanent) within five (5) business days after receipt of such notice and shall approve or disapprove such repairs within ten (10) business days after receipt of such notice.
2. Dust Control. Applicant shall be responsible for dust control on the City roadways, as generated by those vehicles associated with the Project. Dust control shall be performed according to the Weld County Code.

M. Applicant's Proportional Share of Costs. As agreed to by the Parties in this Agreement, the Applicant's Proportional Share shall be Forty Percent (40%). After completion of the Permanent Improvements no other Proportional Share shall be required from the Applicant in regard to maintenance or repair.

N. Annual Road Inspection. The City may conduct a road inspection annually during the Interim Period. As a result of the annual inspection, the City, in its sole but reasonable discretion, shall determine actual conditions and shall further determine what road repair/maintenance work is to be performed during the Interim Period in order to maintain the Haul Route in its Current Condition during that construction season and shall provide written notice of any such

repair/maintenance work required to Applicant no less than thirty (30) days' prior to any such work.

PART III: GENERAL PROVISIONS

O. Permits: Applicant is required to apply for and receive all permits required by the City or any other applicable local, State, or federal permit, including but not limited to:

1. Access Permit. Except for access as provided by that certain City of Evans Access Permit No. 0016-2016, Applicant shall not use any other access onto any City road unless and until an access permit has been issued by the Engineering Division of the City of Evans, in accordance with the Site Plan. Engineering may condition the issuance of an updated access permit on the amendment of this Agreement if the updated access permit authorizes the use of an additional access point, or if there is a change in use of the current access point, as permitted by the Site Plan and/or Evans Municipal Code.
2. Right-of-Way (ROW) Permits. Per Chapter 12.04 of the Evans Municipal Code, any work including grading, occurring within a ROW requires a ROW permit issued by the City Engineer. No work shall occur without said ROW permit.

P. Enforcement and Remedies:

1. Violation of Terms of Agreement. If, in the City Engineer's good faith opinion, Applicant has violated any of the terms of this Agreement, City may notify Applicant of its belief that the Agreement has been violated and shall state with specificity the facts and circumstances which City believes constitute the violation. Applicant shall have thirty (30) days within which to either cure the violation, dispute the violation or demonstrate compliance, unless a longer period is reasonably required in which case Applicant shall be entitled to such longer period so long as it is diligently pursuing a cure of the violation or demonstration of compliance. If, after thirty (30) days or such longer period has elapsed, City believes in good faith that the violation is continuing, City may consider Applicant to be in breach of this Agreement.
2. Remedies for Breach. If the City determines that the Applicant is in breach of this Agreement, and Applicant is not then continuing good faith efforts to cure such breach and has failed to correct said violation, the City may exercise any or all the following remedies, or any other remedy available in law or equity.
 - i. Withholding Permits. The City may "lock" any or all lots within the Property, which precludes the continued issuance of building permits, zoning permits, or any other permit issued by any department of the City.
 - ii. Court Action. The City may seek relief in law or equity by filing an action in the Weld County District Court or Federal District Court for the District of Colorado, except that no such civil action or order shall be necessary to take any other administrative action authorized by the Evans Municipal Code or this Agreement.
 - iii. Revocation of Permit. Applicant acknowledges that failure to comply with the terms of this Agreement constitutes cause to suspend or revoke the Access Permit, and City may exercise this option in its sole discretion by proceeding with suspension or revocation under the current provisions of the Evans Municipal Code. Suspension or revocation of the Access Permit would prohibit the operation of the Project, until such time an Access Permit is reinstated or reissued.

3. Remedies for Breach by the City. If at any time the Applicant determines the City is in breach of this Agreement, the Applicant may provide the City with written notice. In the event the City fails to remedy the alleged breach within thirty (30) days of such written notice, or such additional periods (up to an additional thirty (30) days) as may be reasonably necessary to cure in light of the nature of the alleged breach, provided that the City commences such cure within such thirty (30) day period and continuously prosecutes such cure to completion during such additional period, then the Applicant's remedies are limited to specific performance of this Agreement, subject to the availability, appropriation and budgeting of funds by the City Council, and shall not include any claim for damages or other monetary relief under any circumstances, either in law or equity. Notwithstanding any other provision of this Agreement, in the event the Applicant makes a payment to the City for the Permanent Improvements (whether by the City drawing upon the escrow funds, the bond or other surety as provided in Part II Section K above, or otherwise) but the City fails to substantially complete the Permanent Improvements within the Interim Period or other mutually agreed additional period, the City shall be liable to Applicant for all amounts paid by Applicant to the City subject to this Agreement.

Q. Termination of Agreement. This Agreement shall terminate upon the earliest of the following:

1. Revocation of Underlying Permit. Revocation of the underlying Land Use Permit under Evans Municipal Code. However, the City may still exercise all necessary activities under this Agreement that the City determines necessary to protect the health, safety, and welfare of the residents of Evans.
2. Cessation of all Permit Related Activities. Upon Applicant's non-temporary and complete cessation of all activities permitted by the Access Permit including any clean up or restoration required. A partial cessation of activities shall not constitute a termination of this Agreement, nor shall City's issuance of a partial release/vacation constitute a termination hereof. Unless informed in writing by the Applicant of cessation of activities, and verified by the City, cessation shall only be presumed if the City determines that the operation has been inactive for five (5) years.
3. Non-appropriated and Budgeted Funds. The Parties expressly agree and understand that the City is a "body politic and corporate"/"Home Rule Municipal Corporation" under Colorado law and its ability to perform is contingent upon the future availability of funds which can then be appropriated and budgeted by the City Council. Notwithstanding any other provision of this Agreement, this Agreement will terminate upon the expiration of the Interim Period (including, if applicable, the Tail Period). If appropriated and budgeted funds for the contemplated Permanent Improvements are not fully funded, and construction of the Permanent Improvements is not completed, by the expiration of the Interim Period (or, if the Tail Period applies, by the Tail Period Expiration Date), then this Agreement shall automatically terminate at the expiration of the Interim Period and the City shall refund the Applicant's Proportional Share paid into escrow (including any accrued interest thereon and any unused portion of the maintenance costs contributed by Applicant), or otherwise fully release any payment bond or other surety provided by Applicant, in each case, within five (5) business days following the termination date in accordance with this provision.

R. No Debt; No Multiple-Fiscal-Year Obligation. The Parties acknowledge and agree that this Agreement does not constitute a debt, indebtedness, or multiple-fiscal-year direct or indirect

debt or other financial obligation whatsoever of the City within the meaning of any constitutional, charter, or statutory limitation, including TABOR. The City's obligations are limited to annually appropriated amounts and do not pledge the City's credit or taxing power.

S. General Provisions:

1. Successors and Assigns. This Agreement shall run with the land. However, the Applicant may not delegate, transfer, or assign this Agreement in whole or in part, without the prior express written consent of City, not to be unreasonably withheld, conditioned or delayed, and the written agreement of the party to whom the obligations under this Agreement are assigned. Consent to a delegation or an assignment will not be unreasonably withheld by the City. In such case, Applicant's release of its obligations shall be accomplished by City's execution of a new development agreement with the successor owner of the Property.
2. Severability. If any term or condition of this Agreement shall be held to be invalid, illegal, or unenforceable by a court of competent jurisdiction, this Agreement shall be construed and enforced without such provision, to the extent that this Agreement is then capable of execution within the original intent of the Parties.
3. Governmental Immunity. No term or condition of this Agreement shall be construed or interpreted as a waiver, express or implied, of any of the immunities, rights, benefits, protections or other provisions, of the Colorado Governmental Immunity Act §§24-10-101 et seq., as applicable now or hereafter amended.
4. No Third-Party Beneficiary. It is expressly understood and agreed that the enforcement of the terms and conditions of this Agreement, and all rights of action relating to such enforcement, shall be strictly reserved to the undersigned parties and nothing in this Agreement shall give or allow any claim or right of action whatsoever by any other person not included in this Agreement. It is the express intention of the undersigned parties that any entity other than the undersigned parties receiving services or benefits under this Agreement shall be an incidental beneficiary only.
5. Entire Agreement/Modifications: This Agreement including the Exhibits attached hereto and incorporated herein, contains the entire agreement between the parties with respect to the subject matter contained in this Agreement. This instrument supersedes all prior negotiations, representations, and understandings or agreements with respect to the subject matter contained in this Agreement. This Agreement may be changed or supplemented only by a written instrument signed by both parties.
6. Evans City Council Approval. This Agreement shall not be valid until it has been approved by the Evans City Council, Colorado.
7. Choice of Law/Jurisdiction. Colorado law, and rules and regulations established pursuant thereto, shall be applied in the interpretation, execution, and enforcement of this Agreement. Any provision included or incorporated herein by reference which conflicts with said laws, rules and/or regulations shall be null and void. In the event of a legal dispute between the parties, the parties agree that the Weld County District Court or Federal District Court for the District of Colorado shall have exclusive jurisdiction to resolve said dispute.

8. Attorneys' Fees/Legal Costs. In the event of a dispute between the City and Applicant, concerning this Agreement, the parties agree that each party shall be responsible for the payment of attorney fees and/or legal costs incurred by or on its own behalf.
9. Release of Liability. To the extent the City is not immune or protected as a matter of law under the Colorado Governmental Immunity Act, C.R.S. §§ 24-10-101, et seq., as amended from time to time, Applicant shall indemnify and hold harmless the City from any and all liability loss and damage City may suffer as a result of all suits, actions, or claims of every nature and description to the extent caused by, arising from, or on account of the negligent acts or omissions of Applicant, or Applicant's agents, assigns or employees. The Applicant further agrees that all contractors and other employees engaged by Applicant in construction of the improvements required to be performed by Applicant pursuant to this Agreement shall maintain adequate worker's compensation insurance and public liability insurance coverage and shall operate in strict accordance with the laws and regulations of the State of Colorado governing occupational safety and health.
10. Authority to Sign. Each person signing this Agreement represents and warrants that he or she is duly authorized and has legal capacity to execute and deliver this Agreement. Each party represents and warrants to the other that the execution and delivery of the Agreement and the performance of such party's obligations hereunder have been duly authorized, and that the Agreement is a valid and legal agreement binding on such party and enforceable in accordance with its terms. If requested by the City, the Applicant shall provide the City with proof of the Applicant's authority to enter into this Agreement within five (5) days of receiving such request.
11. Acknowledgment. The Parties acknowledge that each has read this Agreement, understands it and agrees to be bound by its terms. Both parties further agree that this Agreement, with the attached or incorporated Exhibits, is the complete and exclusive statement of agreement between the parties and supersedes all proposals or prior agreements, oral or written, and any other communications between the parties relating to the subject matter of this Agreement.
12. Interpretation. Notwithstanding any other provision contained in this Agreement, for any conflict or potential conflict between any term in this Agreement and the Resolution of the Evans City Council approving the underlying land use permit, the provisions of the Council's Resolution shall control. In the event of a discrepancy between this Agreement and the Evans Municipal Code in effect at the time of the Agreement, the terms of the Evans Municipal Code shall control.
13. Survival. The Parties agree that their respective rights and obligations under this Agreement shall survive the termination or expiration of this Agreement to the extent necessary to protect the rights and interests of each Party or otherwise give proper effect to the provisions hereunder that are intended to survive such termination or expiration.

[SIGNATURES ON NEXT PAGE]

IN WITNESS WHEREOF, the parties hereto each herewith subscribe to the same as of the date first written above.

ATTEST:

CITY OF EVANS, COLORADO

By: _____
Julie Barnett, City Clerk

By: _____
Mark C. Clark, Mayor

Date

APPROVED AS TO FORM:

APPROVED AS TO CONTENT:

By: _____
Edward C. Bull IV, City Attorney

By: _____
Cody R. Sims, City Manager

APPLICANT: MOUNTAIN CEMENT COMPANY LLC

By: _____

Name: _____

Title: _____

STATE OF WYOMING)
)
COUNTY OF ALBANY)

ss.

The foregoing instrument was acknowledged before me this _____ day of August, 2026, by
_____.

WITNESS my hand and official seal.

Notary Public

IN WITNESS WHEREOF, the parties hereto each herewith subscribe to the same as of the date first written above.

ATTEST:

CITY OF EVANS, COLORADO

By: _____
Julie Barnett, City Clerk

By: _____
Mark C. Clark, Mayor

Date

APPROVED AS TO FORM:

APPROVED AS TO CONTENT:

By: _____
Edward C. Bull IV, City Attorney

By: _____
Cody R. Sims, City Manager

APPLICANT: MOUNTAIN CEMENT COMPANY LLC

By: Ben Bufmach

Name: Ben Bufmach

Title: VP operations

STATE OF WYOMING)
COUNTY OF ALBANY)

ss.

The foregoing instrument was acknowledged before me this 12th day of August, 2026, by Ben Bufmach.

WITNESS my hand and official seal.

Ann Prince
Notary Public

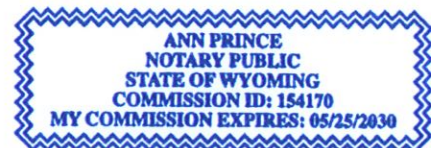


EXHIBIT A – PROPERTY DESCRIPTION

Lot 1 of the Mountain Cement Company Minor Subdivision, being a part of the SE4 of the Section 3, T4N, R66W of the 6th P.M. of Weld County, Colorado, recorded July 25, 2025, at Weld County reception no. 5043502.

EXHIBIT B – SITE PLAN

The Site Plan (case number 24-SP-04), was recorded with the Weld County Clerk and Recorder on the 7th day of July, 2025, at reception no. 5039435.

EXHIBIT C – HAUL ROUTE MAP AND DESCRIPTION

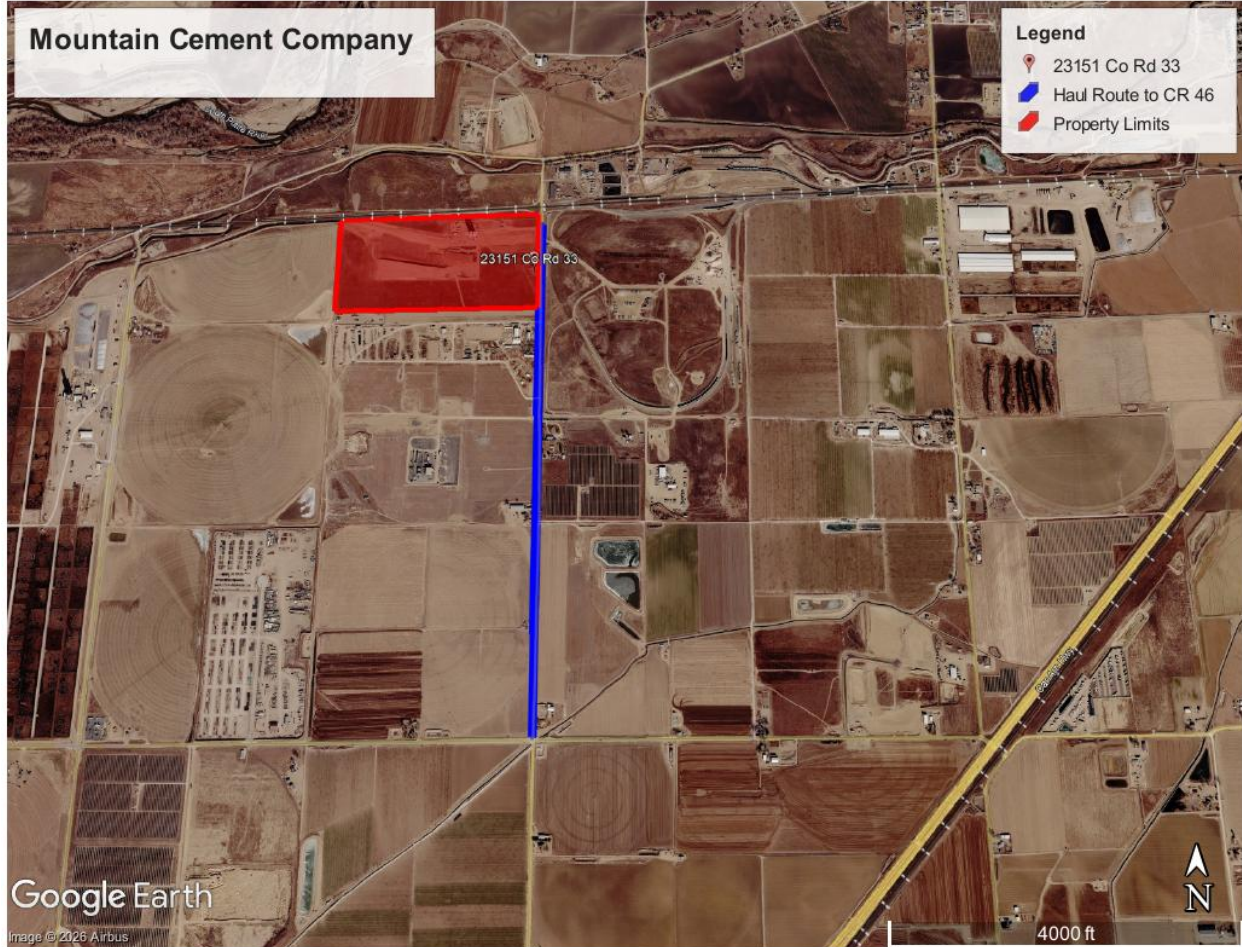


EXHIBIT D

Existing Haul Route Conditions





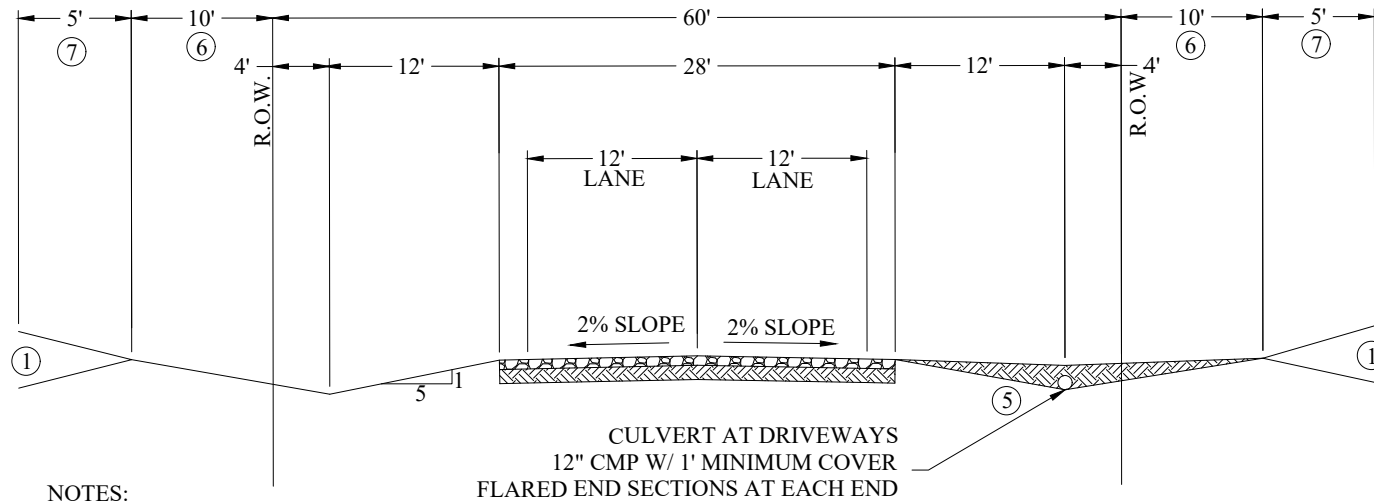




EXHIBIT E

City's Engineering Standards and Specifications

(See attached)



NOTES:

1. CUT AND FILL SLOPES SHALL BE A MAXIMUM OF 4:1.
2. RIGHT OF WAY AND EASEMENT AREAS SHALL BE GRADED (CUT & FILL) TO SUBGRADE $\pm 6"$ AT UTILITY LOCATIONS, INCLUDING SERVICES, PRIOR TO UTILITY INSTALLATION.
3. STANDARD CROWN SLOPE IS 2%. WITH SPECIAL DESIGN REVIEW, 1% TO 4% IS ALLOWABLE AT TRANSITION AND OTHER ATYPICAL SECTIONS.
4. DRAINAGE SWALES SHALL BE CONSTRUCTED ON EACH SIDE OF RIGHT OF WAY, AS SHOWN. MAXIMUM 5:1 SHOULDER SLOPE PERMITTED.
5. DRIVEWAY CROSSING PERMITTED AT APPROVED LOCATIONS ONLY. ALL CROSSINGS SHALL INCLUDE 12" CORRUGATED METAL PIPE (CMP) CULVERT WITH FLARED END SECTIONS ON EACH END. 12" MINIMUM COVER SHALL BE PLACED OVER CULVERT PIPE, AND DRIVEWAY SHALL BE GRADED TO PREVENT SURFACE FLOW DISCHARGE INTO THE STREET. ALL DRIVEWAYS AND CULVERTS ARE TO BE CONSTRUCTED AND MAINTAINED BY THE PROPERTY OWNER.
6. DRAINAGE EASEMENT REQUIRED, 10' MINIMUM.
7. UTILITY EASEMENT REQUIRED, 5' MINIMUM.
8. PERMITTED ONLY FOR RURAL AND LARGE LOT DEVELOPMENTS WITH LOT SIZES OF ONE ACRE OR GREATER.



Public Works
Department

ROADWAY SECTION DETAIL RURAL LOCAL

Scale: N.T.S

Detail No. S-27

May 2019

SECTION 2 ROADWAY STRUCTURAL DESIGN

3.02.01 GENERAL

The design of the pavement cross-section shall be performed by a Professional Engineer registered in the State of Colorado; whose expertise is in geotechnical engineering. The engineer's report for the pavement thickness design and the soils report upon which it is based will be submitted for review and approval prior to the final approval of any construction plans.

The geotechnical engineer shall submit a letter confirming that subgrade in place immediately prior to pavement operations is still in conformance with the pavement design.

Flexible pavements are those pavements which have sufficiently low bending resistance to maintain continuous contact with the underlying structure yet have sufficient stability to support a given traffic loading condition.

Rigid pavements are those pavements which possess a high bending resistance and distribute loads over a large area of foundation soil. Examples include Portland Cement concrete pavement, or Portland Cement concrete surfaced with asphalt.

3.02.02 PAVEMENT DESIGN CRITERIA

The design of streets shall be based on the design period of twenty (20) years. If a single pavement design is used for all streets within the development, that pavement design shall be based on the worst soil encountered from the standpoint of subgrade support.

The design of pavements shall be based upon the Roadway Design Manual of the State of Colorado, Department of Transportation, as modified by these specifications.

The Colorado Department of Transportation methodology is accepted for both rigid and flexible street pavement designs.

The strength coefficients of the various layers of the pavement structure shall be determined from the Colorado Department of Transportation Roadway Design Manual

A. CONCRETE PAVEMENT

The minimum concrete structural section shall be seven (7) inches of non-reinforced Portland Cement concrete pavement placed on six (6) inches of Class 4 or Class 5 base and compacted subgrade. Construction plans for concrete pavement shall include a layout for locations of construction joints. Concrete pavement shall not be allowed on local streets.

If the minimum pavement section is to be used, the Soils Report and Pavement Design must demonstrate the adequacy of the structural section for the soil conditions encountered. The City reserves the right to require concrete or treated subgrade in locations where traffic, utilities, type of construction, or time of construction would make asphalt on aggregate impractical.

B. ASPHALT PAVEMENT

Flyash subgrade treatment shall be required with all new roads constructed in the City of Evans. A pavement design shall be submitted that includes the flyash treatment of the subgrade recommended in these specifications by a Geotechnical Engineer licensed in the state of Colorado.

Full-depth asphalt is not an accepted alternate if treated subgrade is used. When flyash is used, a bond breaker shall be used. The thickness and the type of bond breaker shall be recommended by the geotechnical engineer and approved by the City.

Table 3.02-11
Flexible Pavement Design Criteria

Road Classification	20-Year Design Traffic Information		Serviceability Index (psi)			Reliability	Aggregate Base Thickness Class 5 or 6	Asphalt Thickness (inches)	Minimum Structural Number (3)
	M in. EDLA (2)	M in. ESAL	Si Init	St Final	D psi	%	Layer, inches	Layer, inches	
							Min	Min	
Local									
Local #1 & #2	5	36,500	4.5	2	2.5	65	6	4	2.42
Rural Local	5	36,500	4.5	2	2.5	65	6	4	2.42
Collector									
Minor Collector #1	25	182,500	4.5	2.3	2.2	75	6	5	2.86
Major Collector	50	365,000	4.5	2.3	2.2	85	6	6	3.30
Comm. Collector	50	365,000	4.5	2.3	2.2	85	6	6	3.30
Blvd. Collector	50	365,000	4.5	2.3	2.2	85	6	6	3.30
Arterial									
Gateway Arterial	200	1,460,000	4.5	2.5	2	90	8	6	3.52
Arterial	200	1,460,000	4.5	2.5	2	90	8	6	3.52
Notes									
1	Wearing surface course shall be Grading S or SX for residential roadway classification and Grading S for collector, arterials, and all industrial/commercial roadways.								
2	EDLA shall be calculated based on projected traffic uses. Minimum EDLA values are as given for the design lane. The City Engineer may require greater EDLA values if warranted. The EDLA for a Roundabout shall include the cumulative EDLA for each entry leg. EDLA for arterial/collector intersections shall be for two-way traffic.								
3	Minimum structural numbers are based on subgrade R-value=5 and CDOT Calculations; Mr = 3,025 & Std. Deviation = 0.44. CDOT design calculations shall be used for resilient modulus calculations for all roads.								
4	Single lane refers to a paved surface less than 20 feet wide, including residential alleys.								
5	Min/max lift thicknesses: Grade SX: 1½”/ 2½”, Grade S: – 2”/3”.								
6	The minimum HMA section for composite pavements shall be four (4) inches for Local, five (5) inches for Collectors, and six (6) inches for Arterials. The minimum ABC sections shall be as noted in the table above.								

SECTION 3 EXCAVATION, REMOVALS, AND EMBANKMENT

3.03.01 SCOPE

The work covered by this section includes the furnishing of all labor, equipment, supplies, and materials necessary to perform clearing, grubbing, and removal of objectionable materials from the right-of-way prior to grading operations and placement of embankment to conformity with lines, grades, and typical sections as shown on the plans.

All earthwork shall be performed in compliance with the City's MS4 Permit.

3.03.02 SOIL MATERIALS

Soil materials for roadway construction shall be as recommended in the approved geotechnical soils report. Embankment and fill material shall consist of soil, granular sand, gravel, and cobble material, free from frozen material, organic material, trash, glass, broken concrete, other corrosive, or deleterious material. The developer/contractor shall import approved material as necessary. Embankment and fill material must be stable and have a liquid limit less than forty (40) and a plastic index less than fifteen (15) when tested in accordance with AASHTO T-89 and T-91, respectively.

No material shall have a particle dimension larger than six (6) inches, or two-thirds ($\frac{2}{3}$) of the minimum layer depth, whichever is lesser. Where the subgrade layer is less than six (6) inches the maximum size shall not exceed one-half ($\frac{1}{2}$) the depth of the layer. These size restrictions are contingent upon the material being evenly distributed in finer material such that uniform soil consolidation is achieved. If uniform soil consolidation is not being achieved, the City may reduce the size of materials allowed or change the embankment and fill material requirements.

Where unstable subgrade is encountered, the developer/contractor shall take steps necessary to stabilize the material by techniques such as over-excavation and backfill with approved imported material, use of geotextile fabrics, or other combinations. The developer/contractor shall notify the City of the proposed solution to stabilize the subgrade. If required by the City, the developer's design engineer will make recommendations on stabilization techniques and materials. If tests or observations reveal that material being placed is not of suitable quality and structural value, the developer/contractor shall provide other material as approved by the City.

3.03.03 CLEARING AND GRUBBING

The natural ground surface shall be cleared of all vegetation such as trees, logs, upturned stumps, roots of downed trees, brush, grass, weeds, and all other objectionable materials within the limits of construction. All surface objects, trees, roots, and other protruding obstructions not designated to remain shall be cleared and/or grubbed, except stumps and roots and nonperishable solid objects which shall be removed a minimum of two (2) feet below subgrade or slope of embankments. In streets, stumps and root systems shall be removed to below the two (2) foot level. Except in areas to be excavated, all holes resulting from the removal of obstructions shall be backfilled with suitable material and compacted in accordance with these specifications.

Trees which are to be removed shall be removed in such a manner as not to injure standing trees, plants, and improvements which are to remain. Any object that is designated to remain and is damaged shall be repaired or replaced as directed by the City at the developer/contractor's expense. Before the start of any overlot grading, stock piling of soil, or clearing or grubbing operation, the developer/contractor shall file and obtain approval for a State of Colorado permit for Erosion and Sediment Control. Erosion control measures shall always be installed and functional prior to the start of any land disturbing activities and maintained.

3.03.04 REMOVAL AND DISPOSAL OF MATERIALS

All materials removed shall be disposed of outside of the right-of-way. No accumulation of flammable material shall remain on or adjacent to the right-of-way. The roadway and related work areas shall be left with a neat and

finished appearance. In no case shall the material removed from the right-of-way and stockpiled be left in such a manner as to pose erosion, weed, or other hazard to the public. Stockpiles shall have long sloping sides and in such a manner as to accommodate the use of mechanical mowing equipment and not in excess of four (horizontal) to one (vertical) (4:1).

The developer/contractor shall make all arrangements to obtain permission from property owners for disposal or storage locations outside the limits of the project. The developer/contractor shall legally dispose of all material that is deemed unusable by the City, off the project site prior to project completion.

The developer/contractor shall remove all debris from all sidewalks, curb and gutters, pans, and driveways at the construction site and dispose of the debris in an appropriate location.

3.03.05 REMOVAL OF EXISTING IMPROVEMENTS

A. BITUMINOUS PAVEMENT

Bituminous pavement shall be removed to clean, straight lines and should be perpendicular or parallel to the flow of traffic. The developer/contractor shall remove and properly dispose of pavement and road surfaces as a part of trench excavation. If the limits of the asphalt removal are within three (3) feet of the edge of pavement, the pavement shall be removed and replaced to the edge of the pavement. The width of pavement removed along the trench for the installation of pipe shall not exceed the width of the trench specified by more than one (1) foot of each side of the trench without approval of the City.

The edge of the trench should not be closer than three (3) feet to concrete structures (i.e., curb and gutter, sidewalks, driveways, inlets, etc.). The actual distance shall be dependent upon the characteristics of the soil, the type of equipment that is used for trenching, and the methods used for excavation and backfill. If, in the opinion of the City, concrete structures are endangered by undermining of the structure or settlement, the City may require that the structure be replaced, or special construction methods may be required, or the distance of the trench from the structure will be increased.

The developer/contractor shall use such methods such as cutting, drilling, or others that will ensure the breaking of pavement in a straight line. The face of the remaining pavement shall remain approximately vertical. If the edge is damaged during construction, it shall be re-cut prior to final bituminous patching. All saw-cutting shall not extend past the construction limits by more than three (3) inches and such over-cut shall be sealed in the same manner as cracks, or as directed by the City Engineer.

B. CONCRETE PAVEMENT AND STRUCTURES

Concrete to be removed shall be cut vertically with square edges such that each edge of the finished patch will be parallel or perpendicular to the direction of traffic. The edge for removal will be in a straight line set by a string line, chalk line, or other means to ensure a straight removal line deviating no more than one (1) inch in twenty (20) feet.

Cross pans, alley intersections, and concrete pavement shall be removed to the nearest construction joint or saw-cut to a minimum depth of two (2) inches. Sections shall be saw-cut in straight lines either parallel to the curb or perpendicular to the alignment of the sidewalk or curb. All other sawing shall not extend more than the depth of the cut past the construction limits and shall be sealed in the same manner as joints. If the removed portion falls within less than five (5) feet of a construction joint, cold joint, expansion joint, or edge, crack, or existing patch, the concrete shall be removed to the joint, crack, edge, or patch.

The developer/contractor shall avoid breaking away the edges of the existing pavement or damaging the remaining pavement with heavy construction equipment. The sawing of concrete shall be done carefully, and all damages due to developer/contractor's operations to concrete to remain in place shall

be repaired at the expense of the developer/contractor. The limits of removal of damaged concrete shall be determined in the field by the City or an authorized representative.

On sidewalks, bikeways, curb and gutters, valley gutters, driveways, and alley approaches, the minimum removal section shall be the entire panel, unless otherwise approved by the City Engineer.

Where monolithic curb, gutter, and sidewalk is encountered and only removal of curb and gutter is specified, a straight, full depth saw-cut will be required to separate the curb and gutter from the sidewalk.

Where designated for replacement, removed concrete shall be replaced within three (3) days from the time it was removed (weather permitting). In no case shall the developer/contractor be allowed to remove additional concrete until the three (3) day replacement schedule has been re-established.

In order to allow for forming and patch-back when removing curb and gutter or sidewalk abutting asphalt pavement, the developer/contractor shall remove the adjacent asphalt pavement and base course twenty-four (24) inches wide and six (6) inches deep. Asphalt removal shall be performed prior to concrete forming and then saw-cut for a neat, clean edge immediately prior to asphalt paving/restoration.

If a sidewalk or curb and gutter section is removed and replaced without the removal of the adjacent asphalt, the concrete must be placed such that no voids exist between the concrete and asphalt. The fall between the asphalt and lip of the gutter section must also meet the required cross-section within $\pm 1/4$ inch.

3.03.06 TREE REMOVAL

The Parks Division shall be notified, and their approval obtained prior to excavation of, removal, or trimming of any trees or shrubs within the public rights-of-way.

3.03.07 EXCAVATION

Excavation of all materials shall be performed in conformity to the lines and grades indicated on the drawings. Suitable material removed from the excavations may be used in the formation of embankments and backfilling or any other areas within the right-of-way as permitted by the City. Where material encountered within the limits of the work is considered unsuitable by the City, such material shall be excavated as directed by these standards, the plans, or the City, and replaced with suitable material.

Foundations and the pavement structure shall be placed on original, undisturbed soil or on structural backfill extended to the undisturbed soil. Foundations and the pavement structure shall not be founded on existing fill if encountered at the project site. If existing fill is encountered at the subgrade, the developer/contractor shall excavate to original undisturbed soil and bring the grade to the required elevation with approved material. Existing fill material, if encountered at the site, shall be removed. Existing fill may be stockpiled for reuse in backfills and embankments if it meets the requirements of the specifications. The developer/contractor shall remove unsuitable soil material as directed by the City. The disposal of unsuitable soil material is the responsibility of the developer/contractor.

Rock that is encountered at the site shall be excavated to a minimum depth of six (6) inches below subgrade within the limits of the roadbed.

The developer/contractor shall blend the intersection of cut slopes with the slopes of adjacent natural ground surfaces in a uniform manner. The tops of cut slopes shall be flattened and rounded in accordance with the approved plans.

All excavated material shall be stockpiled in a manner that does not endanger the work or workers, and does not obstruct sidewalks, streets, alleys, or driveways. The work shall be done in a manner that will minimize

interference with traffic and drainage of the street. The developer/contractor at the end of each day shall barricade all excavations and ditch lines, remove excess excavated material from travel ways, and thoroughly clean all streets, alleys, and sidewalks affected by the excavation.

Materials encountered during excavation such as trash, organic, or frozen material, and any other material which is unsatisfactory for use as backfill shall be removed from the site and appropriately disposed of at the developer/contractor's expense. Stones, concrete or asphalt chunks larger than six (6) inches, or frozen material shall be considered unsatisfactory backfill and removed by the developer/contractor, prior to project or phase completion. Frozen material, however, may be thawed out and used later.

3.03.08 EMBANKMENT

This work shall consist of the construction of embankments by depositing, placing, and compacting material of acceptable quality and structural value above the natural ground or other surface in conformance with the lines, grades, and cross-sections shown on the drawings or as established. Before any embankment is placed, clearing, tree removal, sod and topsoil removal over the entire area shall be performed in accordance with these specifications.

When an embankment is to be placed on slopes, it shall be continuously benched in horizontal layers to key into the existing slope. Each layer of the embankment material shall not exceed eight (8) inches in loose depth. The developer/contractor shall thoroughly mix and ensure uniform density and moisture for proper compaction.

Whenever material is encountered that is wet or otherwise unstable and is incapable of supporting structures or the roadbed, the material shall be over-excavated to a depth suitable for construction of a stable subgrade. The developer/contractor shall backfill over-excavated areas with a stabilization material approved by the City. An approved geotextile fabric shall be used where required by the City around the stabilization material and on the subgrade to stabilize the subgrade and prevent fines from migrating into the stabilization material.

Cut areas shall be thoroughly compacted for a depth of eight (8) inches below finished subgrade. Each layer in fill and cut areas shall be prepared, graded, and thoroughly compacted by static roller or vibratory equipment, as detailed in these specifications. The base of fill areas should be scarified to a depth of not less than six (6) inches and compacted to not less than ninety-five (95) percent of maximum density $\pm 2\%$ of optimum moisture content as determined by AASHTO T99 or as recommended by a geotechnical engineer, prior to placement of embankment material. Each layer shall be wetted or aerated, if necessary. No embankment material shall be placed upon soft, spongy, or frozen material or other material, the stability of which is, in the opinion of the City, unsuitable for the placement thereof.

SECTION 4 BACKFILL

3.04.01 SCOPE

This section includes the material and construction specifications for backfill and filter materials.

3.04.02 MATERIAL

A. STRUCTURAL BACKFILL

Structural Backfill shall comply with CDOT Standard Specifications Section 703.08.

Class 1 Structural Backfill shall be used on all bridges, box culverts, or where otherwise specified. In addition, this material shall have a liquid limit not to exceed thirty-five (35) and a plasticity index of not over six (6) when tested in accordance with AASHTO T 89 and T 90, respectively.

Class 2 Structural Backfill shall be composed of suitable materials developed on the project. To be suitable for use under this classification, backfill shall be free of frozen soil, wood, or other organic material. If the material contains rock fragments or other materials that, in the opinion of the City, will be injurious to the structure, the native material shall not be used for backfilling, and the developer/contractor shall be required to furnish Class 1 Structural Backfill material.

B. FLOWABLE-FILL

Flowable-fill meeting the requirements in CDOT Standard Specifications Section 206 and 703, should be used to backfill all open cuts in portions of public right-of-way beneath existing paving, curb, gutter, or sidewalk improvements. Alternate backfill methods for large excavations will be considered on an individual review basis with the City. Flash fill may be used in some location with prior approval of the City Engineer.

The amount of water shall be such that the flowable-fill flows into place properly without excessive segregation.

C. FILTER MATERIAL

Filter material shall consist of free draining sand, gravel, slag, or crushed stone. The grading requirements are set forth in CDOT Standard Specifications Section 703.09 and Table 703-5.

During progress of construction, the developer's design engineer will determine the class of filter material to be used. The selected class will be submitted to the City Engineer for approval.

3.04.03 STRUCTURAL EXCAVATION

Unsuitable foundation material shall be removed and disposed of in a manner acceptable to the City. Unsuitable foundation material which is suitable for embankments and suitable surplus excavated material shall be used in the construction of embankments. Unsuitable material removed below designed elevation shall be replaced with approved material.

Rock, hardpan, or other unyielding material encountered in trenches for culvert pipe or conduit shall be removed below the designed grade for a minimum depth of twelve (12) inches. This extra depth excavation shall be backfilled with loose structural backfill (Class 1) or other approved material. The base of structural backfill shall be scarified to a depth of six (6) inches and compacted to a density of not less than ninety-five (95) percent of maximum density $\pm 2\%$ of optimum moisture content as determined by AASHTO T99 or as recommended by a geotechnical engineer, prior to placement of any structural element or structural backfill. The type of compaction shall be the same as that required for structural backfill (Class 2), as specified below.

3.04.04 STRUCTURAL BACKFILL

Backfill shall consist of approved materials uniformly distributed in layers brought up equally on all sides of the structure. Each layer of backfill shall not exceed twelve (12) inches before compacting to the required density and before successive layers are placed. Structural backfill (Class 1) shall be compacted to a density of not less than ninety-five (95) percent of maximum density $\pm 2\%$ of optimum moisture content as determined by AASHTO T99 or as recommended by a geotechnical engineer.

Required density for structural backfill (Class 2) shall conform to the requirements of these specifications.

Compaction methods that produce horizontal or vertical earth pressures, which may cause excessive displacement or overturning, or may damage structures, shall not be used.

Backfill material shall not be deposited against newly constructed masonry or concrete structures until the concrete has developed a compressive strength of eighty (80) percent of the field compressive strength. Shape the surface of the subgrade under structures such that they are not above or more than one and one-quarter inch ($1\frac{1}{4}$) below the required subgrade elevation. Unless otherwise indicated in the contract or directed by the City, all sheeting and bracing used in making structure excavation shall be removed by the developer/contractor prior to backfilling.

3.04.05 FLOWABLE-FILL

Flowable-fill should be placed on top of a minimum of twelve (12) inches of squeegee material in all utility potholes in the right-of-way. The maximum layer of flowable-fill shall be twelve (12) inches.

Compaction of flowable-fill will not be required. Any damage resulting from placing flowable-fill in lifts that are too thick or from not allowing sufficient cure time between lifts shall be repaired at the developer/contractor's expense.

3.04.06 FILTER MATERIAL AND PLACEMENT

Construction requirements for filter material for subsurface drains shall conform to the applicable requirements of CDOT Standard Specifications Section 206.05.

Filter material shall be placed behind bridge abutments, wing walls, and retaining walls as recommended by the developer's design engineer and as provided in the design/construction plans. Wall drain outlets shall be backed with sacked filter material conforming to the gradation requirements for coarse aggregate No. 3 or No. 4 set forth in CDOT Standard Specifications Table 703-1. Filter material shall be placed in horizontal layers along with and by the same methods specified for structural backfill.

SECTION 5 SUBGRADE AND UNIMPROVED AREA PREPARATION

3.05.01 SCOPE

The work covered by this section concerns the furnishing of all labor, equipment, supplies, and materials needed to perform preparation of subgrade within the public right-of-way.

3.05.02 MATERIALS

A. GEOTEXTILE

Geotextile reinforcement shall be as recommended by an experienced geotechnical engineer retained by the owner, developer or developer/contractor and who is a registered PE in the State of Colorado.

B. FLYASH

This item shall consist of treating the subgrade, existing subbase or existing base by pulverization, adding Class "C" flyash, mixing and compacting of the mixed material to the required density. This item applies to natural ground or embankment and shall be constructed as specified herein and in conformity with the typical sections, lines and grades as shown on the Drawings or as established by the Engineer.

Flyash shall meet ASTM Specification 618, Section 3.2, when sampled and tested in accordance with Sections 4, 6 and 8, unless otherwise shown on the plans. Flyash shall be of the Class "C" designation containing a minimum of twenty-five (25) percent calcium carbonate.

The water used in the stabilized mixture shall be clean, clear, and free of sewage, vegetable matter, oil, acid and alkali. Water known to be potable may be used without testing. All other sources shall be tested in accordance with AASHTO T- 26 and approved by the Materials Engineer.

Flyash shall be stored and handled in closed weatherproof containers until immediately before distribution on the road.

If storage bins are used, they shall be completely enclosed.

If flyash is furnished in trucks, each truck shall have the weight of flyash certified on public scales or the developer/contractor shall place a set of standard platform truck scales or hopper scales at a location approved by the City Engineer.

3.05.03 EQUIPMENT

The machinery, tools and equipment necessary for proper prosecution of the work shall be on the project site prior to the beginning of construction operations.

3.05.04 CONSTRUCTION METHODS

It is the primary purpose of this specification to secure a completed course of treated material which contains a uniform flyash/soil mixture with no loose or segregated areas; has a uniform density and moisture content; is well bound for its full depth; and, has a smooth surface for placing subsequent courses. It shall be the responsibility of the developer/contractor to regulate the sequence of his work; to process a sufficient quantity of material to provide full depth as shown on plans; to use the proper amounts of flyash; to maintain the work; and, to rework the courses as necessary to meet the above requirements.

A. PREPARATION OF SUBGRADE

Before other construction operations are begun, the subgrade shall be graded and shaped to enable the flyash treatment of materials in place, in conformance with the lines, grades, and thickness shown on the plans. Unsuitable soil or materials shall be removed and replaced with acceptable material.

The subgrade shall be firm and able to support, without displacement, the construction equipment and the compaction hereinafter specified. Soft or yielding subgrade shall be corrected and made stable by scarifying, adding flyash, and compacting until it is of uniform stability.

If the developer/contractor elects to use a cutting and pulverizing machine that will remove the subgrade material accurately to the secondary grade and pulverize the material at the same time, he will not be required to expose the secondary grade nor windrow the material. However, the developer/contractor shall be required to roll the subgrade, as directed by the Engineer, before using the pulverizing machine and correct any soft areas that this rolling may reveal. This method will be permitted only where a machine is provided which will ensure that the material is cut uniformly to the proper depth and which has cutters that will plane the secondary grade to a smooth surface over the entire width of the cut. The machine shall be of such design that visible indication is given at all times that the machine is cutting to the proper depth.

B. APPLICATION

The flyash shall be spread by an approved spreader at the rates shown on the plans or as directed by the Engineer. A motor grader shall not be used to spread the flyash.

The flyash shall be distributed at a uniform rate and in such manner as to reduce the scattering of flyash by wind to a minimum. Flyash shall not be applied when wind conditions, in the opinion of the Engineer, are such that blowing flyash becomes objectionable to traffic or adjacent property owners.

C. MIXING

The soil and flyash shall be thoroughly mixed by approved rotary mixers or other approved equipment, and the mixing continued until, in the opinion of the Engineer, a homogeneous, friable mixture of soil and flyash is obtained, free from all clods or lumps. Initial mixing after the addition of flyash will be accomplished dry or with a minimum of water to prevent flyash balls. Water required to achieve the specified moisture content for the mixture should be added after initial mixing.

If the soil flyash mixture contains clods, they shall be reduced in size by raking, blading, disking, harrowing, scarifying or the use of other approved pulverization methods so that when all nonslaking aggregates retained on the No. 4 sieve are removed, the remainder of the material shall meet the following requirements when tested at the field moisture condition or dry by laboratory sieves:

- Minimum Passing one and three-quarter ($1\frac{3}{4}$) inch sieve – 100 percent
- Minimum Passing No. 4 sieve – 60 percent

During final mixing, water shall be added to the materials as directed by the Engineer, until the proper moisture content has been secured. Water shall be added through the pulverizing machine or other method acceptable to the engineer to develop a uniform, controlled rate addition of the needed moisture. Final moisture content of the mix, prior to compaction, shall not exceed the optimum moisture content of the mix by more than two (2) percent nor by less than the optimum by more than four (4) percent. Should the natural moisture content of the soil be above the specified range, aeration of the soil may be required prior to addition of the flyash.

D. COMPACTION

Compaction of the mixture shall begin immediately after final mixing of the flyash and be completed within one (1) hour following addition of flyash and water. The material shall be sprinkled as necessary to maintain the optimum moisture. Compaction of the mixture shall begin at the bottom and shall continue until the entire depth of mixture is uniformly compacted to a specified density.

All non-uniform (too-wet, too dry, or insufficiently treated) areas which appear shall be corrected immediately by scarifying the areas affected, adding or removing material as required and reshaping the area by sprinkling and rolling. The surface of the course shall be maintained in a smooth condition, free from undulations and ruts, until other work is placed thereon, or the work is accepted.

The stabilized section shall be compacted to the extent necessary to provide the density specified below:

Description:

For flyash treated subgrade, existing subbase or existing base that will receive subsequent subbase or base courses	Not less than 95 percent maximum dry density (ASTM D-698)
For flyash treated subbase or base that will receive surface course	Not less than 96 percent maximum dry density (ASTM D-698)

In addition to the requirements specified for density, the full depth of the material shown on the plans shall be compacted to the extent necessary to remain firm and stable under construction equipment. After each section is completed, tests as necessary will be made by the Engineer. If the material fails to meet the density requirements, the Engineer may require it to be reworked as necessary to meet those requirements or require the developer/contractor to change his construction methods to obtain required density on the next section. Throughout this entire operation the shape of the course shall be maintained by blading, and the surface, upon completion, shall be smooth and in conformity with the typical section shown on the plans and to the established lines and grades. Blading should be terminated within two (2) hours after blending of the flyash. Should the material, due to any reason or cause, lose the required stability, density and finish before the next course is placed or the work is accepted, it shall be reprocessed, recompact, and refinished at the sole expense of the developer/contractor. Reprocessing shall follow the same pattern as the initial stabilization, including the addition of flyash.

E. FINISHING, CURING, AND PREPARATION FOR SURFACING

After the final layer or course of the treated subgrade, subbase or base has been compacted, it shall be brought to the required lines and grades in accordance with the typical sections.

The resulting base surface shall be thoroughly rolled with a pneumatic tire roller and "clipped", "skinned" or "tight bladed" by a power grader to a depth of approximately fourteen (14) inch, removing all loosened stabilized material from the section. Recompaction of the loose material should not be attempted. The surface shall then be thoroughly compacted with the pneumatic roller, adding small increments of moisture as needed during rolling. If plus No. 4 aggregate is present in the mixture, one complete coverage of the section with the fat wheel roller shall be made immediately after the "clipping" operation. When directed by the Engineer, surface finishing methods may be varied from this procedure provided a dense, uniform surface is produced. The moisture content of the surface material must be maintained within the specified range during all finishing operations. Surface compaction and finishing shall proceed in such a manner as to produce, in not more than two (2) hours, a smooth, closely knit surface, free of cracks, ridges or loose material conforming to the crown, grade, and line shown on the plans.

After the flyash treated course has been finished as specified herein, the surface shall be protected against rapid drying by any of the following curing methods for a period of not less than three (3) days or until the surface or subsequent courses are placed:

- Maintain in a thorough and continuously moist condition by sprinkling.
- Apply a two (2) inch layer of earth on the completed course and maintain in a moist condition.
- Apply an asphalt membrane to the treated course, immediately after same is completed. The quantity and type of asphalt approved for use by the Engineer shall be sufficient to completely cover and seal the total surface of the base between crown lines and all voids. If the developer/contractor elects to use this method, it shall be the responsibility of the developer/contractor to protect the asphalt membrane from being picked up by traffic by either sanding or dusting the surface of same. The asphalt membrane may remain in place when the proposed surface or other base courses are placed. Asphaltic emulsions are not acceptable for the asphaltic membrane.

3.05.05 SUBGRADE COMPACTION

Field compaction densities for embankments and subgrade soils for improved areas shall be as indicated in the soil report but not less than ninety-five (95) percent of maximum density $\pm 2\%$ of optimum moisture content as determined by AASHTO T99.

Field compaction densities for embankments and soils for unimproved areas shall be not less than ninety (90) percent of maximum density $\pm 2\%$ of optimum moisture content as determined by AASHTO T99.

Do not compact topsoil.

The developer/contractor shall, when requested by the City, furnish the necessary equipment to proof-roll the subgrade even though density tests may indicate compliance with density requirements. Soft and yielding material and other portions of the subgrade which will not compact when rolled or tamped shall be removed as directed by the City and replaced with suitable material. If, based on proof-rolling, the City determines that the subgrade exhibits visible deflection in the form of weaving or pumping and/or the soil cracks, the developer/contractor shall be required to correct the problem to the satisfaction of the City.

Subgrade surfaces below excavated areas such as cut areas and undisturbed areas requiring additional preparation shall be scarified to a minimum depth of six (6) inches, moisture adjusted as needed and compacted until the required density and stability is obtained unless otherwise approved by the City. Scarification shall be done in such a manner as to achieve a uniform mixture of the subgrade soils. Pockets of clay, sand, dry, wet, or stratified soil layers are to be thoroughly mixed, creating a soil blend of uniform consistency.

Scarification shall mean a thorough mixing of the subgrade soils to the required depth either by equipment-mounted ripper teeth or turning the soil with a construction disc. If the scarified soil does not exhibit sufficient subgrade strength for roadway construction, the developer's geotechnical engineer shall be consulted, and alternate methods of construction may be approved by the City. Application of geotextile reinforcement may be accepted if recommended by the developer's geotechnical engineer.

If the required compaction is not obtained, it shall be the responsibility of the developer/contractor to re-compact the material. In cases where there is a failure to achieve the required compaction, the City may require that the backfill be removed and replaced with approved backfill material.

If the developer/contractor's scheduling of work, methods, or inclement weather caused a naturally suitable subgrade to become unstable, the area shall be sub-excavated and stabilized as described above at the developer/contractor's expense.

Acceptable compaction test results shall not relieve the developer/contractor from correction or repairing of any substandard work before or during the two-year (2) warranty period.

3.05.06 GEOTEXTILES

Geotextile reinforcement shall be used in areas of unstable subgrade if recommended by the developer's geotechnical engineer.

3.05.07 SUBGRADE TREATMENT

The surface of the roadbed shall be bladed to the established lines, grades, and cross sections as shown on the plans. The prepared roadbed shall be scarified to the depth and width required for the subgrade stabilization. The material thus obtained shall be pulverized. Application, mixing, and finishing shall be in accordance with CDOT Standard Specifications, Section 307.05 through 307.07. Hydrated Lime shall conform to the requirements of ASTM C 207, Type N. In addition, the residue retained on a #200 mesh sieve shall not exceed ten (10) percent when determined in accordance with ASTM C 110 (drying of the residue in an atmosphere free from carbon dioxide will not be required).

Use of a pozzolan material to treat the subgrade requires a mix design to be submitted by the developer's geotechnical engineer for approval. The pozzolan-treated subgrade will be tested upon completion to verify projected "R" values were achieved.

If any subgrade treatment is placed in the subgrade, a layer of road base shall be placed as a bond breaker before the placement of asphalt.

3.05.08 SUBGRADE SURFACE TOLERANCE

The excavation and embankments for the street, intersections, and driveways shall be finished to a reasonably smooth and uniform surface. Variations from the subgrade shall not exceed more than one-quarter ($\frac{1}{4}$) inch above or one-half ($\frac{1}{2}$) inch below the required subgrade elevation in soil and shall be sub-excavated to a minimum depth of six (6) inches in rock.

3.05.09 COMPACTION IN UTILITY TRENCHES, CULVERTS, ETC.

Utility trenches within the street right-of-way and easements (including service lines) must be mechanically compacted as indicated in the soils report but not less than ninety-five (95) percent of maximum density $\pm 2\%$ of optimum moisture content, as determined by AASHTO T99. All water and sewer services, including water and sewer main stub-outs, shall be installed prior to street construction. Compaction of service trenches shall extend to the street right-of-way line as a minimum. Water settlement and/or jetting of trenches shall not be permitted.

SECTION 6 SUB-BASE

3.06.01 GENERAL

The sub-base, when required for stabilization of the subgrade, shall consist of a foundation course composed of granular material. The sub-base shall be placed on the prepared subgrade in accordance with these specifications and in conformance with the lines and grades and typical cross sections as shown on the plans or established.

3.06.02 MATERIALS

Sub-base material must meet the limits set forth in the CDOT Standard Specifications Table 703-2 for Class 1 material.

The material supplied shall be a well graded mixture, consisting of sound aggregate particles and sufficient filler or other proper quality binding material which, when placed and compacted, will result in a firm, dense, unyielding foundation. Balls of clay within the graded mixture will not be accepted.

3.06.03 PLACEMENT AND COMPACTION

Each layer of sub-base material shall be placed in layers not to exceed twelve (12) inches in compacted depth. Each layer shall be wetted or aerated, if necessary, and compacted to ninety-five (95) percent maximum density $\pm 2\%$ percent of optimum moisture as determined by AASHTO T99 or as recommended by a Geotechnical Engineer. No sub-base material shall be placed upon a soft, spongy, frozen, or otherwise unacceptable subgrade or other unsatisfactory subgrade, the stability of which is unsuitable for the placement thereof.

3.06.04 SUB-BASE SURFACE TOLERANCE

The prepared surface of the sub-base shall not vary by more than one-quarter ($\frac{1}{4}$) inch above or one-half ($\frac{1}{2}$) inch below the required subgrade elevation.

SECTION 7 BASE COURSE

3.07.01 GENERAL

The base shall consist of a foundation course composed of crushed gravel or crushed stone and filler, constructed on the prepared sub-base or subgrade in accordance with these specifications and in conformance with the lines and grades as shown on the plans or established. This work shall be completed in accordance with CDOT Standard Specifications Section 304 except as modified herein.

3.07.02 MATERIALS

A. BASE COURSE MATERIAL

A design mix shall be submitted to the City for review and approval. Aggregates for bases shall be crushed stone, crushed slag, crushed gravel, natural gravel, or crushed reclaimed concrete or asphalt material which conforms to the quality requirements of AASHTO M 147 except that the requirements for the ratio of minus #200 sieve fraction to the minus #40 sieve fraction, stated in 2.2.2 of AASHTO M 147, shall not apply. The requirements for the Los Angeles abrasion test (AASHTO T 96) do not apply to Class 1, 2, and 3 aggregates.

Crushed gravel or crushed stone base course material shall consist of hard durable particles or fragments of stone or gravel crushed to required size and a filler of sand or other finely divided mineral matter. The portion of the material retained on a No. 4 sieve shall be known as coarse aggregate, and that portion passing a No. 4 sieve shall be known as filler. When produced from gravel, not less than sixty (60) percent by weight of the coarse aggregate particles, shall be particles having at least one fractured face, and if necessary, to meet this requirement, or to eliminate an excess of filler, the gravel shall be screened.

The composite base course material shall be free from vegetation and lumps or balls of clay and shall meet the grading requirements for the class specified for the project. The liquid limit shall be per CDOT Standard Specifications Table 703-2 in accordance with AASHTO T 89, and the plasticity index shall not exceed six (6) when the aggregate is tested in accordance with AASHTO T 90.

The Soils Report shall identify areas that in the developer's geotechnical engineer's opinion need to be treated with mineral filler or hydrated lime.

For approved aggregate base course classifications see CDOT Standard Specifications Table 703-2.

B. COMMERCIAL MINERAL FILLERS

Portland cement shall conform to CDOT Standard Specifications subsection 701.01. Hydrated lime shall conform to CDOT Standard Specifications subsection 712.03.

C. PLANT MIXED-BITUMINOUS BASE COURSE

Plant mixed bituminous base may be substituted for the combination of asphalt and untreated base on an equivalent strength basis. This material consists of aggregate and bituminous material mixed in a central plant, spread and compacted on a prepared surface as specified in the CDOT Standard Specifications Section 301.

3.07.03 PLACEMENT AND COMPACTION OF BASE COURSE

A. PLACING

If the required compacted depth of the aggregate base course exceeds six (6) inches, it shall be constructed in two (2) or more layers of approximately equal thickness. The maximum compacted thickness of anyone (1) layer shall not exceed six (6) inches. When vibratory or other types of special compacting equipment are used, the compacted depth of a single layer may be increased to eight (8) inches upon request, provided that specified density is achieved and is approved by the City Engineer.

B. MIXING

The developer/contractor shall mix the aggregate by methods that produce a thorough and homogenous mixture.

C. SHAPING AND COMPACTION

Compaction of each layer shall continue until a density of not less than ninety-five (95) percent of the maximum density $\pm 2\%$ of optimum moisture content as determined by AASHTO T180 has been achieved or as recommended by a geotechnical engineer. The surface of each layer shall be maintained during the compaction operations so that uniform texture is produced and the aggregates are firmly keyed. Water shall be uniformly applied during compaction in the quantity necessary for proper consolidation of the material, or the material shall be harrowed, disked, bladed, or otherwise worked to ensure a uniform moisture content.

The subgrade and base course shall be free from standing water during construction. Remove any water encountered during construction to the extent necessary to provide a firm and stable subgrade and base course. Divert surface runoff or use other means necessary to accomplish the above. Do not deposit, tamp, roll or otherwise mechanically compact the aggregate base course in water. Do not construct aggregate base course with frozen material or on frozen subgrade.

The prepared surface upon which the surface course is to be placed will be tested with a ten (10) foot straight edge, or another approved device. The surface shall be tested prior to the application of any primer or pavement. The variation above or below the testing edge of the straight edge between any two (2) contacts with the surface shall not exceed one-quarter ($\frac{1}{4}$) inch. Any areas not complying with these tolerances shall be reworked to obtain conformity.

The in-place compacted thickness of aggregate base course shall be no more than one-quarter ($\frac{1}{4}$) inch less than the thickness shown on the approved drawings.

Immediately prior to paving, the aggregate base course shall be proof-rolled as described in these specifications to verify the base course stability. Areas that are found to be unstable shall be removed and replaced.

3.07.04 TESTING OF AGGREGATE

The developer/contractor shall sample and test the subgrade, sub-base, and base material prior to being used on the project site. Written results of the tests will be submitted to the City within ten (10) working days after the results of the tests are known.

The developer/contractor is responsible for the quality control testing and protection of work until the work is accepted by the City. The developer/contractor will provide acceptance testing. Acceptance testing may include but is not limited to tests associated with the base course subgrade preparation and trench compaction. See the Material Sampling and Testing Schedule in Appendix B for additional requirements.

The City may conduct additional testing. The developer/contractor shall provide access to allow for the City's testing representative, his operations, and materials.

SECTION 8 ASPHALTIC PAVING

3.08.01 GENERAL

These specifications include general requirements that are applicable to all types of plant-mix bituminous pavements, irrespective of gradation of aggregate, kind and quantity of bituminous material, or pavement use.

Paving consists of one (1) or more courses of bituminous mixture constructed on a prepared foundation in accordance with these specifications and the specific requirements of the type proposed for the project, and in conformity with the lines, grades, thicknesses, and typical cross-sections shown on the plans or established. This work shall be completed in accordance with CDOT Standard Specifications Division 400 except as modified herein.

The design engineer shall recommend the appropriate mix designations for a specific pavement design based upon:

- Traffic analysis
- Serviceability
- Drainage/environmental considerations
- Evaluation of the subgrade, sub-base, base, and surface course materials

Reclaimed Asphalt Pavement (RAP) material will be accepted in bituminous plant mixes provided that all the requirements for hot bituminous pavement are met, except that the mix shall not contain more than twenty (20) percent RAP material.

3.08.02 MIX DESIGN

A. GENERAL

The bituminous plant-mix shall be composed of a mixture of aggregate, filler, or additives (if required and approved), bituminous material, and reclaimed material if permitted.

Mix designs can either be general or site specific. All general mix designs shall be in accordance with CDOT Section 411 (asphalt materials) and Section 702 (bituminous materials).

Approved CDOT designations shall be Grading S (three-quarter ($\frac{3}{4}$) inch maximum) for new street construction. Grading SX (one-half ($\frac{1}{2}$) inch maximum) may be used for overlay or, in special cases, as authorized by the City. Grading SG (one-and-one-half inch by seven-eighths ($1\frac{1}{2} \times \frac{7}{8}$) may be used as a bottom lift of asphalt on newly constructed streets.

B. SITE SPECIFIC

The developer/contractor proposed job-mix formula for each hot bituminous pavement grading will be tested utilizing materials actually produced and stockpiled for use on the project.

The job-mix formula for each mixture shall establish a single percentage of aggregate passing each required sieve size, a single percentage of bituminous material to be added to the aggregate, and a single temperature for the mixture at the discharge point of the plant.

Should a change in sources of materials be made, a new job-mix formula shall be established before the new material is used. This new job-mix formula shall be in effect until modified by the City. Requests made in writing by the developer/contractor for changes in the job-mix formula will be considered. A change in the job-mix formula will require a revised mix design to be furnished to the City.

C. QUALITY CONTROL

Quality control of the production of the mixture shall be the sole responsibility of the developer/contractor. All field and laboratory testing shall be performed by a qualified laboratory acceptable to the City. All test results shall be submitted to the City upon completion of the testing.

The City shall have the authority to perform additional acceptance testing at the City's discretion. The cost of re-testing due to failed acceptance shall be borne by the developer/contractor. The developer/contractor shall cooperate with the quality assurance personnel representing the City without delay and shall make the work and materials available for testing at all times.

Typically, quality control testing during production of the mix typically is as shown in Quality Control Test Table, Table 3.08-12.

Table 3.08-12
Quality Control Test

TEST	METHOD	FREQUENCY
Gradation	AASHTO T27	a minimum of one for each mix design
Asphalt Content	AASHTO T164, ASTM D4125	a minimum of one for each mix design
Hveem Stability	AASHTO T246	a minimum of one for each mix design
Lottman	CPL 5109	a minimum of one for each mix design
Rice Value and VMA	Asphalt Institute Methods	a minimum of one for each mix design
Asphalt Oil Certificate of Compliance	---	one from each asphalt oil supplier

Where mixes contain polymerized asphalt, gradation samples shall be obtained from the cold feed. Where mixes contain AC-10F, samples shall be obtained at the construction site at appropriate locations. Where mix gradations exceed the tolerances shown for two consecutive tests, a new mix design shall be formulated prior to proceeding with any work.

3.08.03 SUBMITTALS

The developer/contractor shall submit the following to the City:

A. GENERAL MIX DESIGN

If the mix design conforms to CDOT Standard(s), a certificate of conformance shall be provided to the City.

B. SITE-SPECIFIC MIX DESIGN

No paving shall begin prior to the submittal and approval by the City of such mix designs. Prior to beginning paving each calendar year, the developer/contractor shall submit to the City for review and approval a mix design for each mix.

If a site-specific mix design is proposed, one (1) mix design shall be submitted to the City for each aggregate source and for each pavement type intended for use in constructing public streets.

Aggregate source mix design shall be submitted to the City and shall include the gradation and percentage by weight of each element proposed.

C. SUPPLIER

The name, address, and contact information for the plant supplying asphalt.

3.08.04 MATERIALS

A. AGGREGATES FOR PLANT-MIX HOT BITUMINOUS PAVEMENT

Aggregates for plant-mix hot bituminous pavement shall be of uniform quality, composed of clean, hard, durable particles of crushed stone, crushed gravel, natural gravel, or crushed slag. Excess of fine material shall be wasted before crushing. The natural sand content shall not exceed twenty (20) percent of the weight of the total aggregate blend for Gradings S and SX and fifteen (15) percent for Grading G. Natural sand is unprocessed, naturally occurring fine aggregate composed mostly of round particles. All aggregates shall be non-plastic when tested in accordance with AASHTO T 90.

Reclaimed material shall be of uniform quality. The maximum size of the reclaimed asphalt pavement shall be three-quarter ($\frac{3}{4}$) inch prior to introduction into the mixer.

The material shall not contain clay balls, organic matter, or other deleterious substances. The aggregate for Gradings S and SX shall have a percentage of wear of forty-five (45) percent or less when tested in accordance with AASHTO T 96.

B. MINERAL FILLER

Mineral filler shall conform to the requirements of AASHTO M 17 and shall consist of rock dust, slag dust, hydrated lime, hydraulic cement, flyash, or other suitable mineral matter. It shall be free of organic impurities and agglomerations. When used, it shall be dry enough to flow freely. Mineral filler shall be graded according to CDOT Standard Specifications Section 703.06. Mineral filler shall have a plasticity index not greater than four (4) excluding hydrated lime and hydraulic cement.

C. HYDRATED LIME

As per CDOT Standard Specification Section 712.03.

D. BITUMINOUS MATERIALS

Bituminous materials for paving shall be per CDOT Standard Specifications Section 702.01, meeting performance graded bitumen PG 64-22, unless otherwise approved by the City.

The City may require the developer/contractor to take a sample of the material for testing at any time.

3.08.05 WEATHER LIMITATIONS

Weather limitations shall be as per CDOT Standard Specification Section 401.07 and Section 407.04.

When it is in the public interest and approved by the City, minimum temperature requirements may be waived for placing tack coats and layers of bituminous mixtures below the top layer of the completed pavement. However, pavement operations will be suspended if density requirements are not met.

3.08.06 ASPHALT MIXING PLANT

The bituminous mixing plant shall be capable of producing a uniform material, have adequate capacity, and be maintained in good mechanical condition. Defective parts shall be replaced or repaired immediately if they adversely affect the proper functioning of the plant or plant units, or adversely affect the quality of the hot bituminous plant-mix.

Acceptable safety equipment, approved by the City, shall be provided by the developer/contractor or the supplier to accommodate sampling and testing.

Hot bituminous plant-mix shall not be stored longer than nine (9) hours, unless additional protective measures are used and pre-approved by the City.

When hot bituminous plant mix is obtained from a commercial plant, the developer/contractor shall make arrangements for approved laboratory facilities at the plant site for testing hot bituminous paving mixtures. The plant laboratory shall meet the requirements of CDOT Standard Specifications subsection 620.04.

3.08.07 HAULING EQUIPMENT

Trucks used for hauling bituminous mixtures shall have tight, clean, smooth metal beds thinly coated with a minimum amount of paraffin oil, lime solution, or other approved release agent. Petroleum distillates such as kerosene or fuel oil will not be permitted. Each truck shall have a cover of canvas or other suitable material to protect the mixture from the weather.

Any hauling and/or paving equipment that is leaking oil, fuel, or any other type of liquid, will be removed from the project and will be required to be inspected by the City before it will be allowed back on the project. If the piece of equipment is not fixed and continues to leak on the project, the City may reject the asphalt and require the asphalt to be removed at the developer/contractor's expense.

3.08.08 ASPHALT PAVERS

Self-propelled asphalt pavers shall be provided and equipped with an activated/vibrating screed assembly, heated, capable of spreading and finishing the bituminous plant-mix material in lane widths applicable to the typical section and thicknesses given for the project. Pavers used for shoulders and similar construction shall be capable of spreading and finishing courses of asphalt plant-mix material in widths given for the project.

The paver's receiving hopper shall have sufficient capacity for a uniform spreading operation and shall have an automatic distribution system that will place the mixture uniformly in front of the screed.

The screed or strike-off assembly shall produce the specified finished surface without tearing, shoving, rippling, or gouging the mixture.

The pavers shall be equipped with automatic screed controls with sensors capable of sensing grade from an outside reference line and maintaining the screed at the specified longitudinal grade and transverse slope. The sensor shall be constructed to operate from either or both sides of the paver and shall be capable of working with a ski-type device at least thirty (30) feet in length, or a short ski or short shoe.

The type of devices to be furnished shall be as provided in the specifications for the project.

The controls shall be capable of maintaining the screed at the specified transverse slope within $\pm 0.1\%$.

Manual operation will be permitted for constructing irregularly shaped and minor areas.

If the automatic controls fail or malfunction, the equipment may be operated manually for the remainder of the normal working day, provided specified results are obtained.

If the developer/contractor fails to obtain and maintain the surface tolerances as specified in these specifications, the paving operations shall be suspended until satisfactory corrections, repairs, or equipment replacements are made.

Placement of hot bituminous pavement on a waterproofed bridge deck shall be accomplished with equipment that will not damage the membrane or protective covering.

3.08.09 TACK COAT

Whenever new asphaltic pavement is placed on existing bituminous pavement, or against existing concrete surfaces (including gutter pans, curbs, manholes, and valve boxes) a bituminous tack coat shall be applied to the existing pavement prior to placing the new pavement. When the new pavement abuts the old pavement, the developer/contractor shall cut the old pavement as described in the Street Cut and Excavation Repair section of these Specifications, and spray the edge of the old pavement with a tack coat. The tack coat material shall conform to requirements for bituminous materials found in CDOT Standard Specifications Section 407, latest edition.

The surface to receive the tack coat shall be dry and cleaned by an approved method until all dust, debris, and foreign matter are removed. The tack coat material shall be applied at a rate of between 0.05 and 0.10 gal/yd² and at a temperature which will provide a very thin coating uniformly distributed over the entire area to be covered. The tack coat shall not be applied at temperatures outside the requirement in CDOT Standard Specifications Standard 407.04. At the City's direction, the developer/contractor shall provide adequate means of calibrating or measuring the rate of application which should not exceed 0.10 gal./yd².

3.08.10 SURFACE CONDITIONING

Before starting the paving, the developer/contractor shall ensure that utility lines, piping, general grading, and heavy trucking are complete so such operations will not damage paving work.

Prior to placing the pavement, the developer/contractor shall adjust manholes, valve boxes, and other fixtures to below pavement grade. These fixtures shall be adjusted to final grade after paving. See Standard Details (S-20, S-36 and S-37).

Irregularities in the existing pavement or base shall be brought to uniform grade and cross section. Prior to placing tack coat and beginning overlay work, the surface shall be swept to remove accumulations of loose gravel and debris in those areas where this is applicable. Contact surfaces of curbing, gutters, manholes, and other structures shall be sprayed with a uniform coating of bituminous material prior to placing asphalt mixture against them.

3.08.11 PREPARATION OF ASPHALT CEMENT

The bituminous material shall be heated to the specified temperature without local overheating and shall be continuously supplied to the mixer at a uniform temperature within the specified range.

3.08.12 PREPARATION OF AGGREGATES

Heating and drying of the aggregates shall be accomplished without damaging the aggregate.

When hydrated lime is used it shall be added to the aggregate in accordance with one of the following methods:

A. Lime Slurry Added to Aggregate

The hydrated lime shall be added to the aggregate in the form of slurry and then thoroughly mixed in an approved pugmill. The slurry shall contain a minimum of seventy (70) percent water by weight.

B. Dry Lime Added to Wet Aggregate

The dry hydrated lime shall be added to wet aggregate (a minimum of three (3) percent above saturated surface dry) and then thoroughly mixed in an approved pugmill.

The lime-aggregate mixture may be fed directly into the hot plant after mixing or it may be stockpiled for not more than ninety (90) days before introduction into the plant for mixing with the bituminous material. The hydrated lime may be added to different sized aggregates and stockpiled, by adding seventy-five (75) percent of the lime to the aggregate passing the No. 4 sieve and twenty-five (25) percent to the aggregate retained on the No. 4 sieve.

3.08.13 BITUMINOUS PAVEMENT MIXING

Mixing shall be in accordance with CDOT Specification 401.15 unless otherwise approved by the City.

3.08.14 SPREADING AND FINISHING

Spreading and finishing of hot bituminous pavement shall be in accordance with CDOT Specification 401.16, except as noted herein.

A. Asphalt Pavers and Other Equipment

Asphalt pavers shall be used to distribute the mixture to the established grade and required thickness over the entire width or partial width as practicable.

The developer/contractor shall protect the asphalt (both existing and new) from solvents and oils. Any piece of equipment leaking any fluid shall be removed from the work site immediately and shall not return to the work site until all leaks are repaired. If any piece of equipment leaks any fluid a second time, it shall be removed from the work site immediately and shall not be allowed on the work site again for the remainder of the project. The developer/contractor shall not use diesel or other solvents to remove or prevent the sticking of asphalt to the wheels of rubber-tired rollers or other equipment used on the asphalt.

B. Courses/Layers

The asphaltic surfacing shall be placed in at least two lifts on all street types.

The first lift shall be a minimum (not an average) of two (2) inches finished thickness on arterials and collectors. On local streets, the first lift (or base course) shall be a minimum of one and one-half (1½) inches and a maximum of two (2) inches. The top lift (or wear course) shall be a minimum of two (2) inch finished thickness. A tack coat between lifts is required if the top lift is not immediately placed on the first lift.

C. Surface Tolerance

For surface courses, the finished surface of the bituminous pavement when tested with a ten (10) foot straight-edge parallel to the centerline or perpendicular across joints will show variations as measured from the testing face of the straight-edge to the surface of the pavement, which shall not exceed three-sixteenths (3/16) of an inch or as per CDOT Specification Section 401.20. There shall be no allowance for standing water on the final pavement surface. After compaction, the asphalt pavement adjoining the curb and gutter shall be between one-quarter (¼) inch and one-half (½) inch higher than the lip of the gutter pan. Areas that do not meet the required surface accuracy shall be clearly marked, and, if the City requires repair, the developer/contractor shall correct the area by removing defective work and replacing it with new material as directed.

D. Thickness Tolerance

Any deficiency in the total thickness of the asphaltic pavement shall not exceed ten (10) percent for any one (1) sample with the average deficiency for all samples not to exceed seven and one-half (7½) percent. Final decision for correction of deficiencies shall not be made until a pavement evaluation is made by an independent testing laboratory retained by the developer/contractor.

E. Longitudinal Joints

The longitudinal joint in both new pavement and overlay pavement layers shall offset the joint in the layer immediately below by six (6) inches. The joints in any pavement layer shall not fall in a wheel track. The joints in the top layer of new pavement not built on top of an existing pavement shall be located at the centerline of the pavement and at the outside edge of the travel lanes for two lane roadways and at the lane lines and at the outside edge of the travel lanes for roadways of more than two lanes.

Where paving operations are on the present traveled roadway, the developer/contractor shall arrange paving operations so there will be no exposed longitudinal joints between adjacent travel lanes at the end of a day's run. With the approval of the City, the developer/contractor may be permitted a vertical or tapered exposed longitudinal joint when the thickness of pavement course being placed is one (1) inch or less, or a tapered exposed longitudinal joint when the thickness of pavement course being placed is greater than one (1) inch. Minimum width of taper shall be three times the thickness of the pavement course.

On areas where the use of mechanical spreading and finishing equipment is impracticable, the mixture shall be dumped, spread, raked, screeded, and luted by hand tools to the required compacted thickness. All oversized aggregate accumulating during hand raking shall be removed from the roadway surface and disposed of properly.

Production of the mixture shall be maintained so pavers can be used in echelon to place the wearing course in adjacent lanes.

F. Segregation of Asphalt

The bituminous mixture shall be transported and placed on the roadway without segregation. All segregated areas behind the paver shall be removed immediately upon discovery. The segregated material shall be replaced with specification material before the initial rolling has taken place. If more than fifty (50) square feet of segregated pavement is ordered removed and replaced in any continuous 500 linear feet of paver width lay-down, operations shall be discontinued until the source of the segregation has been found and corrected.

The City will determine the extent of segregated areas. The developer/contractor shall correct all segregated areas at his own cost.

3.08.15 HAND PLACEMENT

In narrow, deep, or irregular sections; intersections; turnouts; or driveways where it is impractical to spread and finish the mixtures by machine methods, the developer/contractor may use acceptable hand methods as accepted by the City.

When the mixture is to be spread by hand, upon arrival on the work site, it shall be dumped on the area on which it is to be spread. Immediately thereafter it shall be distributed into place by means of hot shovels and spread with hot rakes in a loose layer of uniform density and correct depth. Tines of the rakes shall be not less than one-half ($\frac{1}{2}$) inch longer than the maximum diameter of aggregate particles and in no case should the space be less than one (1) inch. Loads shall not be dumped any faster than they can be properly handled by the raker.

The raker will not be permitted to stand in the hot mixture while raking it, except where necessary to correct errors in the first raking. The raking must be carefully and skillfully done in such a manner that after the first passage of the roller over the raked mixture, a minimum amount of back patching will be required.

Placing mixture shall be as continuous as possible and the roller shall pass over the unprotected edge of the freshly laid mixture only when the laying of this course is to be discontinued for such intervals of time as to permit the mixture to become chilled.

In locations where vibratory plate compactor is required/used for hand placed materials, the existing/adjoining pavement shall be saw-cut and removed a minimum of one and one-half (1½) times the plate width to achieve compaction.

3.08.16 COMPACTION

The plant-mix bituminous pavement shall be compacted by rolling. Both steel wheel and pneumatic tire rollers will be required. The number, weight, and type of rollers furnished shall be that which is sufficient to obtain the required density while the mixture is in a workable condition. Compaction shall begin immediately after the mixture is placed and be continuous until the required density is obtained. When the mixture surface temperature falls below 185° Fahrenheit, no further compaction effort will be permitted unless approved.

All roller marks, ripples, or other surface defects shall be removed with the finish rolling. Use of vibratory rollers with the vibrator on will not be permitted during surface course final rolling and will not be permitted on any rolling on bridge decks covered with waterproofing membrane.

Pavement shall be compacted to a density of ninety-two (92) percent to ninety-six (96) percent of the maximum theoretical density, determined according to AASHTO T 209. Along forms, curbs, headers, walls, and all other places not accessible to the rollers, the mixture shall be thoroughly compacted with mechanical tampers.

Any mixture that becomes loose and broken, mixed with dirt, or is in any way defective, shall be immediately removed and replaced with fresh hot mixture, and compacted to conform with the surrounding area.

3.08.17 JOINTS

Placing of the bituminous paving shall be continuous, and rollers shall not pass over the unprotected end of a freshly laid mixture. Transverse joints shall be formed by cutting back on the previous run to expose the full depth of the course. A coat of bituminous material shall be applied to contact surfaces of all joints just before additional mixture is placed against the previously compacted material. Location and configuration of longitudinal joints shall be in accordance with CDOT Standard Specification subsection 401.16.

3.08.18 QUALITY CONTROL

A. SITE SPECIFIC MIX DESIGN

If a site-specific mix design is used, the City may obtain samples from the hot bituminous pavement supplier for analysis by an independent laboratory.

B. TESTING AGGREGATE SPECIMENS

Tests for cleanliness, abrasion loss, and percent of fractured faces will be made on representative samples of aggregate taken during production or from the stockpiles. These tests are the responsibility of the developer/contractor.

C. TESTING ASPHALT SPECIMENS

Production sample testing shall be done in accordance with Materials Sampling and Testing Schedule found in Appendix B.

The developer/contractor shall furnish the bituminous paving mixture necessary for quality control testing. A qualified testing laboratory acceptable to the City shall sample and test the specimens. All

test results shall be submitted to the City within ten (10) working days of the completion of testing. The developer/contractor shall be responsible for the quality control tests.

The primary purpose for sampling and testing fresh bituminous paving mixtures is to determine conformity with the job-mix formula for acceptance. Results of these tests may be used by the producer to control uniformity of the mixture at the place of manufacture. It is, therefore, very important that the samples of bituminous mixture represent, as nearly as possible, what the manufacturer is producing. A typical quality control testing program is as follows:

- One density test per 1,000 linear feet of pavement per day with a minimum of three (3) tests per project.
- Obtain one (1) sample for each 500 tons or fraction thereof of mix produced on a project, but not less than one (1) sample per project.
- One (1) Marshall/Rice test and gradation and extraction test per day of paving operation, unless the paving for the day is less than one hundred (100) tons.

3.08.19 ACCEPTANCE TESTING

The developer/contractor will provide production sample testing and acceptance testing.

Cores will be taken at the City's discretion to determine specific density, depth thickness of the asphaltic pavement, and compliance with the design mix. Core samples should be taken after the final lift, or as directed by the City. Where the core samples have been taken, new material shall be placed and compacted into the holes by the developer/contractor to conform with the surrounding areas.

Testing the in-place pavement for compliance with the requirements for surface smoothness may be required by the City. The City may perform additional acceptance testing if it is necessary to determine that the pavement is acceptable or to determine the extent of unacceptable pavement. The developer/contractor shall repair or remove and replace any unacceptable pavement as required by the City.

The developer/contractor shall be responsible for the costs associated with re-testing due to failed acceptance tests. The developer/contractor may contract with a qualified engineer to evaluate the unacceptable pavement and propose a remedy for review and approval by the City.